IEOR 4404: Simulation

Lecture 13: Variance Reduction and Importance Sampling

Simulation Output Analysis

Recall simulation output analysis:

- Suppose we are interested in estimating $\mu = E[g(X)]$
- Point estimate

  $\hat{\mu}$ = average of simulation replications of $g(X)$

- How do we quantify uncertainty?

To simplify, we want to estimate $\mu = E[X]$.

from $X_1, \ldots, X_n$ (data or simulation replications)

we calculate $\hat{\mu} = \frac{X_1 + \cdots + X_n}{n}$

Sample Size vs Estimation Variance

Suppose the variance of $g(X)$ is σ^2

measure of uncertainty

Half-width of confidence interval = $z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$

(assuming n is large enough and σ is known)

To achieve a margin of error ϵ, we can:

- Increase sample size n (required sample size $\approx \left(z_{1-\frac{\alpha}{2}} \frac{\sigma}{\epsilon}\right)^2$)

- Decrease variance σ^2

decrease σ

$z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$

shrink half-width

increase n

Sample Size vs Estimation Variance

More concretely, if we can lower the variance σ^2, then the required sample size is smaller

In fact, the required sample size is proportional to σ^2

Suppose we want

$$\text{half-width} = Z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \varepsilon,$$

if we can lower σ, then we can use a smaller n. $\Rightarrow$ a faster simulation to achieve the same accuracy.

Sample Size vs Estimation Variance

An alternative perspective:

Recall the Chebyshev inequality in the proof of the Law of Large Numbers:

$$P(|\hat{\mu} - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

To ensure a large error, i.e., $|\hat{\mu} - \mu| > \epsilon$, happens only with a small probability, say α, we need

$$\frac{\sigma^2}{n\epsilon^2} \leq \alpha$$

or equivalently

$$n \geq \frac{\sigma^2}{\alpha\epsilon^2}$$

If σ is lowered, the required n such that $P(|\hat{\mu}-\mu| > \epsilon) \leq \alpha$ is also lowered.

Again, the required sample size n is proportional to σ^2

Variance Reduction

To summarize, to improve estimation accuracy, we can either increase sample size or decrease variance

In some cases, it is more economical, or even necessary, to decrease variance instead of increasing sample size

Examples:

- Each simulation run is expensive
- Rare-event problems, where variance is low but the target accuracy is very high

estimation of probabilities that are tiny

Variance Reduction

Methods to reduce σ^2:

- Importance sampling
- Control variates

most popular

- Conditional MC
- Stratification
- Common random numbers

also popular

- Antithetic variables ← limited effect
- Multilevel MC
- Quasi MC

powerful but can be difficult to use

Importance Sampling

Suppose we want to estimate $E[g(X)]$, where X has a density f

naive Monte Carlo

Naïve MC: Generate X_i from f and output

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i)$$

IS idea:

- Use a different density $\tilde{f}$ to generate X
- Multiply by a "likelihood ratio" to de-bias the estimate

Importance Sampling

Intuitively,

- Using a different density to generate X is an artificial adjustment to the target problem $\Rightarrow$ It leads to a wrong, or in other words biased, estimate

- By multiplying with the likelihood ratio, the estimate is "de-biased"

- The hope is to use $\tilde{f}$ that leads to a lower ultimate variance

Importance Sampling

- Suppose we generate X from $\tilde{f}$

  different from f

- Define

  original density

$$L(X) = \frac{f(X)}{\tilde{f}(X)} \text{ as the likelihood ratio}$$

  IS density

- With n simulation runs, we output

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i)L(X_i)$$

Importance Sampling

Claim: Each IS run, $g(X)L(X)$, where $X \sim \tilde{f}$, is unbiased

Suppose $X \sim \tilde{f}$. Then $\tilde{E}[g(X)L(X)] = E[g(X)]$

Reasoning:

$$E[g(X)] = \int g(x)f(x)dx = \int g(x)\frac{f(x)}{\tilde{f}(x)}\tilde{f}(x)dx$$

$$= \tilde{E}[g(X)L(X)]$$

A "change of measure" idea

Importance Sampling

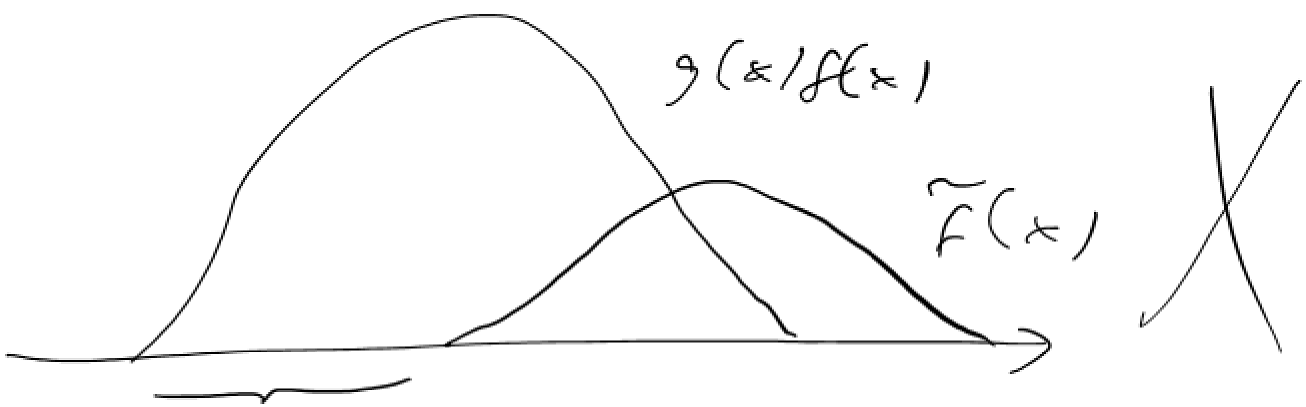
Only makes sense if $g(x)\frac{f(x)}{\tilde{f}(x)}$ is well-defined

Requirement: $g(x)f(x)$ has support at most that of $\tilde{f}(x)$

i.e., if $\tilde{f}(x) = 0$, then $g(x)f(x) = 0$, or
if $g(x)f(x) \neq 0$, then $\tilde{f}(x) \neq 0$

Example: To estimate $P(X > 10)$ for $X \sim N(0, 1)$, can we use $\tilde{f} =$

- $N(20, 20)$
- $Uniform(-20, 20)$
- $Exp(1)$



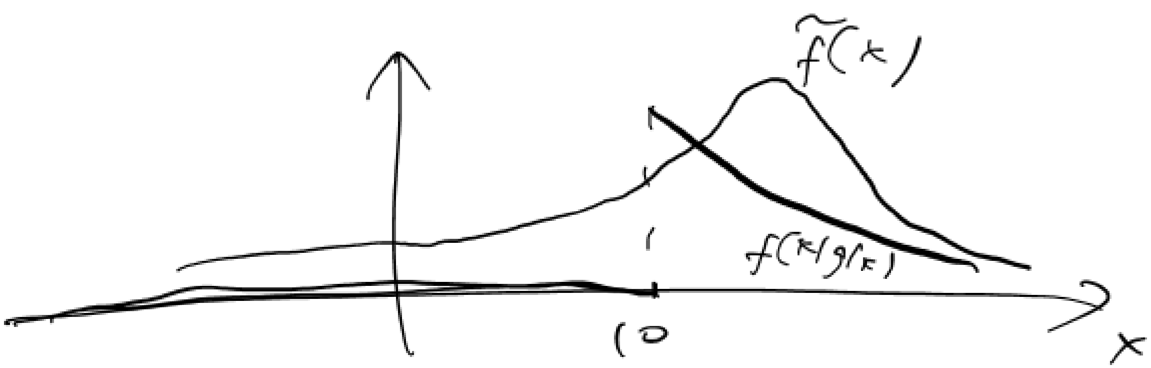
$g(x)f(x) > 0$ but $\tilde{f}(x) = 0$

Target : $P(X>10) = E[I(X>10)]$

$g(x) = I(x>10)$

$f(x) = N(0,1)$

$\tilde{f}(x) = N(20,20)$

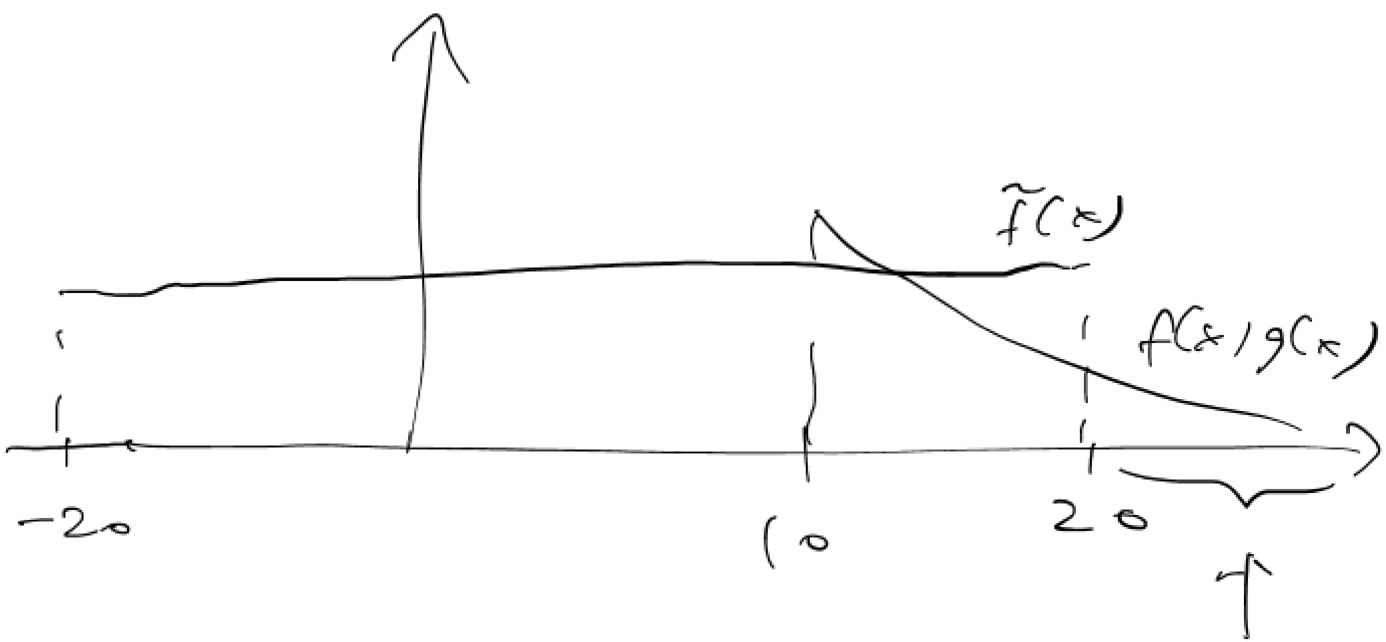


whenever $\tilde{f}(x)=0$,

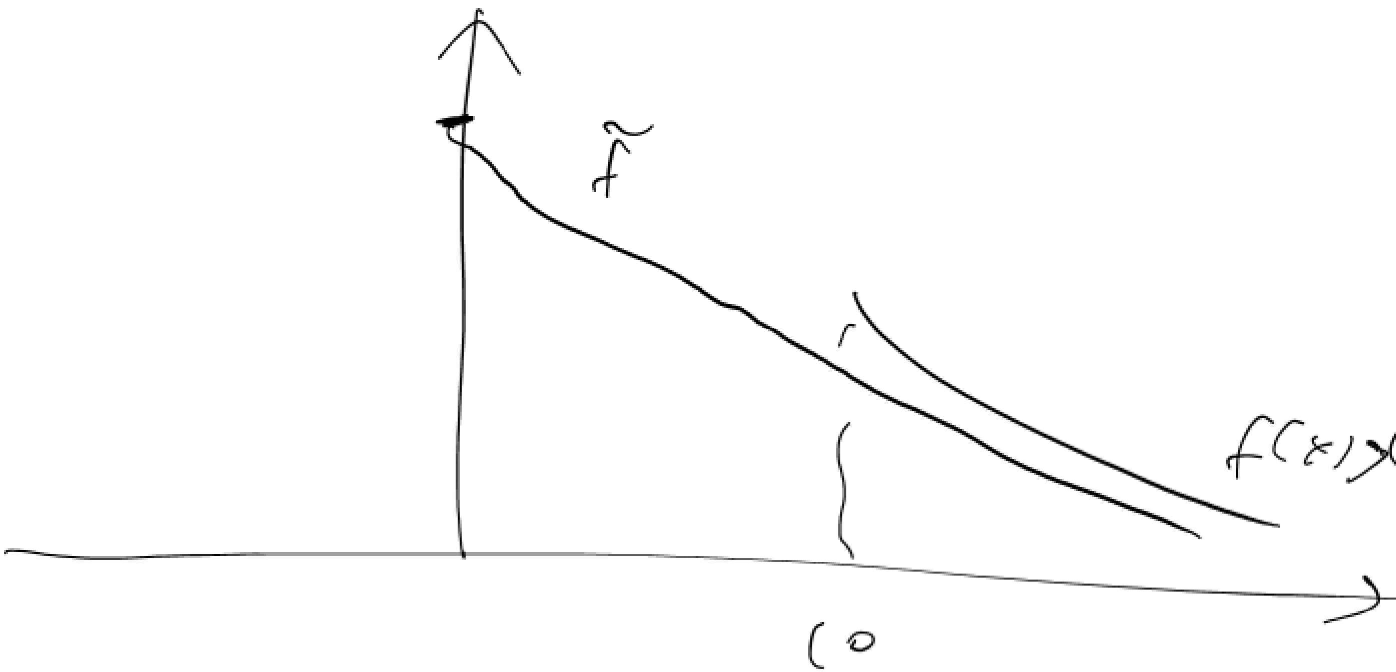
$f(x)g(x)=0$

Obviously because

$\tilde{f}(x) > 0$ for all x



when $\tilde{f}(x) = 0$, we can have $f(x)g(x) > 0$.



whenever $\tilde{f}(x)=0$, $f(x)g(x)=0$.

Rare-Event Simulation

In rare-event probability estimation, our goal is to estimate target quantity p = P(rare-event)

What is a meaningful estimate?

- Suppose the true p (typically unknown) is 10^{-6}
- Suppose we simply use 0 as the estimate. It might appear that we are doing very well, since the error is $|10^{-6} - 0| = 10^{-6}$ which is very small
- However, the estimate 0 is uninformative, as it doesn't tell us how small the probability is. In fact, the goal is usually to try to get the magnitude on how small is the probability of the rare event

Rare-Event Simulation

Why is rare-event simulation difficult?

- Suppose we use naïve Monte Carlo, i.e., generate many $X_i \sim f$ and output

$$\frac{1}{n}\sum_{i=1}^{n} I(X_i > 10)$$

  to estimate $P(X > 10)$.

- It would take a very large replication size to observe one "hit" of $X_i > 10$, since this event is rare. For a reasonable n, we would have all $I(X_i > 10) = 0$ and so we output 0, the uninformative estimate

Rare-Event Simulation

$|\hat{p} - p| \leq \varepsilon p \Rightarrow$ meaningful estimate $\hat{p}$

- To capture our intuition of what accuracy level is meaningful, we should view $|\hat{p} - p| > \epsilon p$, not merely $|\hat{p} - p| > \epsilon$, as a large-error situation

- Note that $\hat{p} = 0$ is considered to have a large error

- In other words, we want a point estimate $\hat{p}$ to be close to p relative to the magnitude of p

Rare-Event Simulation

- To understand how many sample size is needed to obtain an estimate with a good relative discrepancy, by Markov or Chebyshev inequality,

$$P(|\hat{p}-p|>\epsilon p)\le\frac{\sigma^2}{n\epsilon^2 p^2}$$

required n so that

$P(|\hat{p}-p|>\epsilon p)\le\alpha$

- The needed n to achieve a relative discrepancy of ϵ with confidence $1-\alpha$ is $\ge\frac{\sigma^2}{\alpha\epsilon^2p^2}$

$\frac{\sigma^2}{n\epsilon^2p^2}\le\alpha$

- The quantity $\frac{variance}{mean^2}=\frac{\sigma^2}{p^2}$ is called the relative error. The higher the relative error, the larger n is needed

In rare-event estimation, the relative error $\frac{\sigma^2}{p^2}$ is the criterion for being a good estimator.

Rare-Event Simulation

Why is rare-event simulation difficult (more mathematically understood)?

$$\frac{1}{n}\sum_{i=1}^{n} \underbrace{I(X_i > 10)}_{\sim Ber(p)}$$

Relative error of naïve Monte Carlo

$$= \frac{\sigma^2}{p^2} = \frac{p(1-p)}{p^2} = \frac{1-p}{p}$$

which blows up when $p \approx 0$

A Rare-Event Example

Consider simulating $P(X > 10)$ where $X \sim N(0,1)$

- Naive Monte Carlo: run n simulation runs and obtain

$$\frac{1}{n}\sum_{i=1}^{n} I(X_i > 10)$$

- Suppose we want to estimate p within 5% of the truth, with confidence 95%, then we need a sample size to be

$$\frac{\sigma^2}{5\% \times \ 5\%^2 \times p^2}$$

$$n \geq \frac{\sigma^2}{\alpha \varepsilon^2 p^2}$$

5% 5%

where $\sigma^2 = p(1-p)$

- This number turns out to be around 10^{26}
- If we simulate 1 million normal variables in a millisecond, we need 3 million years to finish

A Rare-Event Example

Consider an IS distribution $\tilde{f} = N(\gamma, 1)$

Likelihood ratio: $L(x) = \frac{f(x)}{\tilde{f}(x)} = \frac{e^{-\frac{x^2}{2}}}{e^{-\frac{(x-\gamma)^2}{2}}} = e^{-x\gamma + \frac{\gamma^2}{2}}$

Relative error:

$$RE^2 = \frac{\widetilde{Var}(I(X > \gamma)L)}{p^2} = \frac{\left(\tilde{E}[I(X > \gamma)^2 L^2] - p^2\right)}{p^2} = \frac{\tilde{E}[I(X > \gamma)^2 L^2]}{p^2} - 1$$

where second moment of IS:

$$\tilde{E}[I(X > \gamma)^2 L^2] = \tilde{E}[L^2; X > \gamma] = \tilde{E}\left[e^{-2x\gamma + \gamma^2}; X > \gamma\right]$$
$$= e^{-\gamma^2} \tilde{E}\left[e^{-2\gamma(x-\gamma)}; X > \gamma\right] \leq e^{-\gamma^2}$$

So

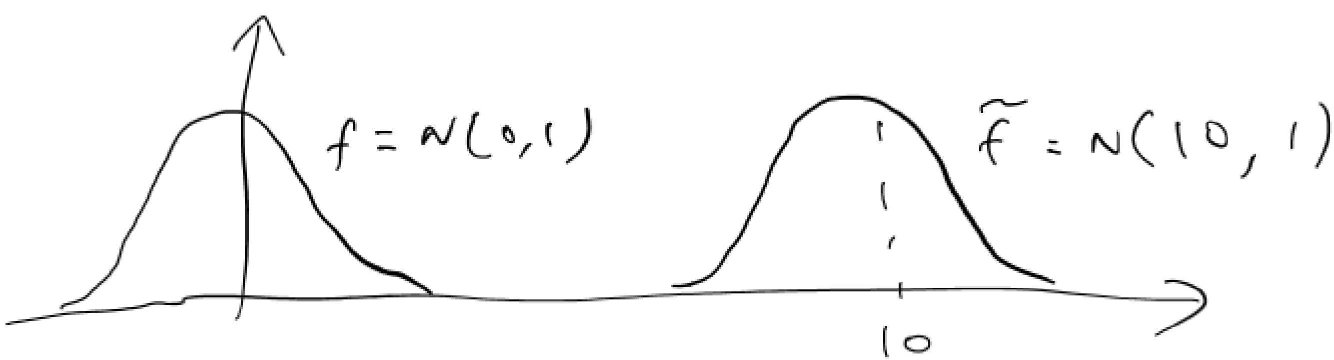
$RE^2 \leq O(\gamma^2) - 1$, polynomial in $\gamma \Rightarrow$ substantial improvement over naïve MC

Goal: Estimate $P(X > 10)$ where $X \sim N(0,1)$

- Naive MC is bad... relative error $= \frac{1-p}{p}$

  which blows up as $p \approx 0$.

- Alternative: use importance sampling with $\tilde{f} = N(10,1)$



Intuitively, by picking $\tilde{f}$ to be $N(10, 1)$, we increase the frequency of hitting $X > 10$. But we need to make sure the likelihood ratio is controlled.

relative error $= \frac{\sigma^2}{p^2} = \frac{\widetilde{var}(I(X>10)\,L(X))}{P(X>10)^2}$

<u>Note:</u> Importance sampling method uses

$$\frac{1}{n}\sum_{i=1}^{n} I(X_i>10)\,L(X_i)$$

where $L(x) = \frac{f(x)}{\tilde{f}(x)} = \frac{\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}}{\frac{1}{\sqrt{2\pi}}e^{-\frac{(x-10)^2}{2}}} = e^{-10x+\frac{10^2}{2}}$

Let $\gamma = 10$.

$$\text{relative error} = \frac{\widetilde{\mathrm{Var}}\left(I(x>\gamma)\, L(x)\right)}{P(x>\gamma)^2}$$

$$= \frac{\tilde{E}\left[\left(I(x>\gamma)\, L(x)\right)^2\right] - \Big(\overbrace{\tilde{E}\left[I(x>\gamma)\, L(x)\right]}^{\substack{\text{since IS is unbiased} \\ P(x>\gamma) \\ \parallel}}\Big)^2}{P(x>\gamma)^2}$$

$$= \frac{\tilde{E}\left[\left(I(x>\gamma)\, L(x)\right)^2\right]}{P(x>\gamma)^2} - 1$$

To analyze the first term,

$P(X > \gamma)$ for $X \sim N(0,1)$

$$\approx \frac{1}{\sqrt{2\pi}\,\gamma} e^{-\frac{\gamma^2}{2}}$$ (true but needs to be proved)

when γ is large.

$$\tilde{E}\left[\left(I(X>\gamma)L(X)\right)^2\right] = \tilde{E}\left[L(X)^2 I(X>\gamma)\right]$$

$$= \tilde{E}\left[e^{-2\gamma X + \gamma^2} I(X>\gamma)\right] = e^{-\gamma^2}\tilde{E}\left[e^{-2\gamma(X-\gamma)} I(X>\gamma)\right]$$

But

$$e^{-\gamma^2} \tilde{E}\left[e^{-2\gamma(x-\gamma)} \; I(x > \gamma) \right]$$

$$\le e^{-\gamma^2}$$ because if $x > \gamma$, then $x - \gamma > 0$.

and $e^{-2\gamma(x-\gamma)} < 1$

relative error $\le \dfrac{e^{-\gamma^2}}{\left(\frac{1}{\sqrt{2\pi}\,\gamma} e^{-\frac{\gamma^2}{2}} \right)^2} \simeq 2\pi \gamma^2$

This is the relative error of IS.

relative error of naive MC

$$= \frac{1-p}{p} \approx \frac{1}{p} \approx \frac{1}{\frac{1}{\sqrt{2\pi}\gamma} e^{-\frac{\gamma^2}{2}}} \approx \sqrt{2\pi}\,\gamma\, e^{\frac{\gamma^2}{2}}$$

<u>Conclusion:</u>

naive MC, $RE \approx e^{\frac{\gamma^2}{2}}$ (super-exponential in γ)

IS, $RE \approx \gamma^2$ (polynomial in γ)

IS is much more efficient.

Exponential Tilting

A convenient framework to design IS

Assume f has light tail, i.e., the logarithmic moment generating function $\psi(\theta) = \log E\left[e^{\theta X}\right]$ exists for θ in a neighborhood of 0.

Consider an exponential family $\tilde{f}_\theta(x) = e^{\theta x - \psi(\theta)} f(x)$

Example:

- $P = N(\mu, \Sigma) \rightarrow \widetilde{P_\theta} = N(\tilde{\mu}, \Sigma)$
- $P = \mathrm{Exp}(\lambda) \rightarrow \widetilde{P_\theta} = Exp(\tilde{\lambda})$

moment generating function (MGF) of a random variable X is

$$E[e^{\theta X}]$$

$$\log \text{MGF} = \log E[e^{\theta X}]$$

Not all r.v. has a valid MGF, because the expectation $E[e^{\theta X}] = \infty$. E.g. X has density $\frac{1}{(1+x)^{\alpha}}$

Define

$$\tilde{f}_\theta(x) = e^{\theta x - \psi(\theta)} f(x)$$

This is called exponential tilting

· $\tilde{f}$ is a valid density, because $\tilde{f}_\theta \geq 0$ and

$$\int_{-\infty}^{\infty} \tilde{f}_\theta(x)\,dx = 1 \quad \left(\int_{-\infty}^{\infty} e^{\theta x - \psi(\theta)} f(x)\,dx \right.$$

$$\left. = e^{-\psi(\theta)} \int_{-\infty}^{\infty} e^{\theta x} f(x)\,dx = \underbrace{e^{-\psi(\theta)} e^{\psi(\theta)}}_{=1} \right)$$

Example:

$f(x)$ is a normal density $N(0,1)$

The exponential tilting of $f(x)$:

MGF of $N(0,1) = e^{\frac{\theta^2}{2}}$

log MGF of $N(0,1) = \frac{\theta^2}{2}$

$$\tilde{f}_\theta(x) = e^{\theta x - \frac{\theta^2}{2}} f(x) = e^{\theta x - \frac{\theta^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}$$

$\Rightarrow$ exponential tilting of $N(0, 1)$

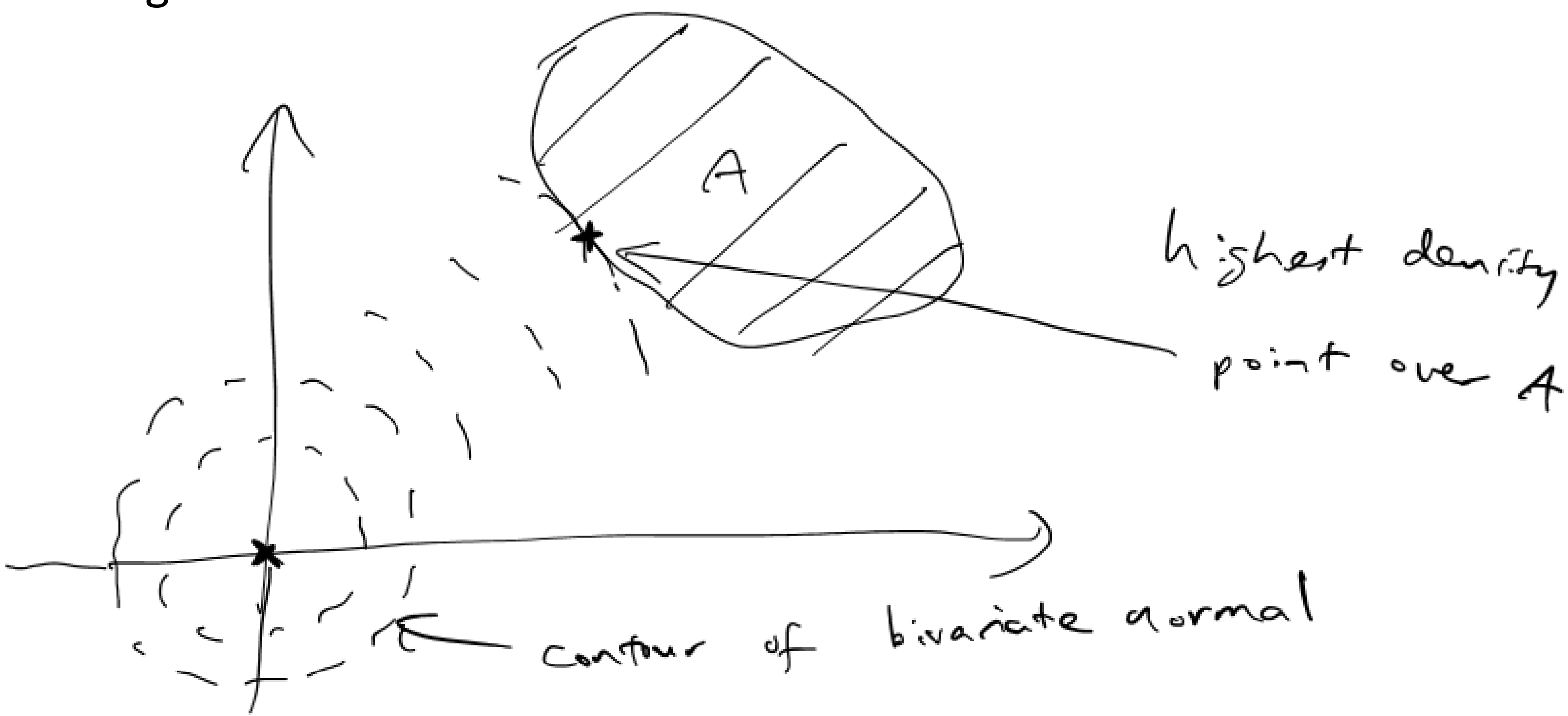
is $N(\theta, 1)$

Exponential tilting

To design IS to estimate $P(X \in A)$ for some set A:

- Move the mean of X to the highest point of density in A, by using exponential tilting

- Generally speaking, this $\tilde{f}_\theta$ would increase the frequency of hitting A, while controlling the variance



Example

Estimate $P(X > 10)$ for $X \sim N(0,1)$, by using an exponential tilting on X so that the mean is moved to the highest density in the target region

For $N(0,1)$,

exponential tilting $= N(\theta, 1)$

By choosing $\theta = 10$, the mean of

the exponential tilting density is 10.

$\Rightarrow$ This can be used as an importance

sampling density.